Embedding and Clustering Persian Prompts the UniRoute Way

A review of the K-means stage of UniRoute on public Persian data with Jina v5 embeddings

Metis-Auto v2 — work package WP-03b

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Abstract

UniRoute (Jitkrittum et al., ICLR 2026) routes each prompt to a large language model (LLM) by clustering prompt embeddings with K-means and describing every LLM by its error rate per cluster. Before spending money on model answers for our Persian gateway, we tested the first half of that pipeline, *embedding and clustering*, on public Persian data. We follow the paper's protocol exactly, replacing the Gecko embedder with Jina v5 (*small* and *nano*). We use 70,186 unique texts from twelve sources: nine labelled benchmarks, among them the 20,577-question Khayyam Challenge, and 10,651 real Persian chat messages. All numbers are measured on held-out test prompts over ten random splits with 95 % confidence intervals. The study made no API calls and cost $0. Four findings stand out. (i) The paper's qualitative result carries over to Persian. A mix of benchmarks forms clear clusters (NMI ≈ 0.35 against nine sources); a single topic forms almost none (NMI ≈ 0.10). Real chat lies in between. (ii) Jina *small* and *nano* are equivalent within 0.015 NMI; nano is better separated and about 40 % faster. (iii) Clusters are meaningful: on Khayyam they gather subjects far above chance (NMI 0.41 vs. 0.005 shuffled). On chat, five of eight clusters carry a clear topic (coding and technology, business, history and society, health, science). (iv) Treating human examinees as one "model", clusters shrink the per-question error gap that bounds UniRoute's loss by 5–6 % on Khayyam and by 0 % on a single topic. With 400 validation prompts the gain peaks near $K = 8$–16, so validation size, not the embedder, limits K. We discuss what this means for the embedder choice, the choice of K and the spec's cluster-size rule.

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1 Introduction

Metis-Auto is a gateway that sends each Persian chat message to one of four candidate LLMs. Its router follows UniRoute [1]. Prompts are embedded and grouped by K-means. Each LLM is then described by its error rate in every cluster, measured on a small labelled validation set. A new prompt goes to the LLM with the lowest "cluster error + λ·cost". Because LLMs are represented by these per-cluster profiles, new models can be added without retraining.

The router is only as good as its clusters. If the clusters do not separate prompts that different LLMs handle differently, every LLM looks the same in every cluster, and routing reduces to always picking one model. The paper shows exactly this contrast. On EmbedLLM, a mix of many benchmarks, K-means routing is strong. On Headlines, where every prompt is financial news, it is weak.

The project's real dataset has not arrived yet, and the rules forbid paid calls before it does. We therefore asked whether the clustering stage works on Persian at all, using only free public data. This report answers five questions:

RQ1 Do Jina v5 embeddings plus K-means recover known topics in Persian text, and how does this depend on K?

RQ2 Does the paper's "benchmark mix vs. single topic" contrast appear in Persian data?

RQ3 Which embedder variant should the project prefer for clustering: *small* or *nano*, and the `clustering` or the `text-matching` adapter?

RQ4 Are the clusters meaningful, i.e. were prompts grouped correctly, including on unlabelled chat?

RQ5 Would these clusters help routing, and what limits the useful number of clusters?

Scope. This report covers embedding, clustering and cluster assignment (spec stages S4–S5). The later stages need a right/wrong label per LLM per prompt: per-model profiles, the choice of K by the area under the deferral curve, and routing itself. Public Persian data has no such labels, so they are left for later work. §4.3 uses a human-difficulty signal as a stand-in.

2 Background

2.1 UniRoute with K-means

Let $\varphi : \mathcal{X} \to \mathbb{R}^D$ be a frozen text embedder with unit-norm outputs. UniRoute's cluster-based variant has four steps:

1. **Cluster** a large unlabelled training set S_{tr} with K-means into centroids $\mu_1, \ldots, \mu_K$. A prompt's cluster is
$$k(x) = \arg\max_k \ \varphi(x)^\top \mu_k. \tag{1}$$
2. **Profile** each LLM h on a small labelled validation set S_{val} (about 400 prompts):
$$\Psi_k(h) = \frac{1}{|C_k|} \sum_{x \in C_k} \mathbf{1}[h(x) \neq y(x)], \qquad C_k = \{x \in S_{\text{val}} : k(x) = k\}. \tag{2}$$
3. **Route** a new prompt to
$$r(x) = \arg\min_h \ \Psi_{k(x)}(h) + \lambda\, c(h), \tag{3}$$
where $c(h)$ is the cost of model h and $\lambda \geq 0$ trades quality for cost.
4. **Choose** K from $\{3, \ldots, N_{\text{val}}/50\}$ by the largest area under the deferral curve (accuracy against cost as λ varies), computed with the training LLMs (App. G.1 of [1]).

With $K = 1$ the rule picks the same LLM for every prompt ("ZeroRouter"). The paper's Proposition 2 bounds the excess loss of cluster routing by how far a prompt's error lies from its cluster's average error. Clusters help when they gather prompts of similar difficulty *for each model*.

2.2 Embedders

The paper uses Gecko 1B (768 dimensions). We use **Jina embeddings v5 text**, which the project chose in decision D-048 for its multilingual quality and local use: *small* (677 M parameters, 1,024 dimensions) and *nano* (768 dimensions, about 5× faster on our machine). Jina v5 has task adapters (LoRA). We use the `clustering` adapter, merged into the weights, as configured in `configs/pipeline.yaml`. The `text-matching` adapter, chosen for the semantic cache, serves as a reference. An earlier benchmark (WP-03) found that only *nano* meets the gateway's 60 ms latency budget at 512 tokens.

3 Data

3.1 Sources

We downloaded twelve public Hugging Face datasets (Table 1). Nine are labelled benchmarks; three are collections of real chat conversations, from which we keep only the first user message, as in the project's spec (D-045). Khayyam is gated; its authors granted access. Texts are deduplicated exactly and empty texts dropped. Only questions and messages are embedded, never answer choices or replies. The agent running the experiments never read any text. All analysis uses counts, labels and scores, and texts appear only on local, git-ignored pages for the author.

Table 1: Sources. "Kept" = unique non-empty texts; "Labels" / "Fine" = number of coarse / fine classes (– = none); length in characters (median / 90th percentile); "Persian" = share tagged Persian. PerCoR, legal QA and medical QA are seeded samples.

Source	Kind	Rows	Kept	Labels	Fine	Chars p50/p90	Persian	Licence
khayyam	benchmark	20,805	20,577	38	95	115 / 314	91%	cc-by-nc-nd-4.0
global_mmlu	benchmark	14,042	13,867	6	57	102 / 641	99%	apache-2.0
persian_arc	benchmark	7,787	7,778	–	2	89 / 204	100%	cc-by-sa-4.0
lmsys_fa	chat	5,431	5,401	–	–	52 / 124	97%	mit
persian_bbh	benchmark	4,511	4,511	–	19	345 / 738	94%	unknown
percor	benchmark	4,000	3,998	–	43	108 / 194	100%	apache-2.0
wildchat	chat	3,913	3,222	–	–	59 / 611	89%	odc-by
legal_qa	benchmark	3,000	3,000	–	–	283 / 666	100%	cc
med_qa	benchmark	3,000	2,822	–	–	45 / 65	99%	apache-2.0
parsinlu_mc	benchmark	2,460	2,436	3	3	53 / 125	100%	unknown
arena	chat	2,160	2,028	–	2	71 / 528	96%	cc-by-4.0
include	benchmark	548	546	5	18	75 / 235	100%	apache-2.0
Total			70,186				96%	

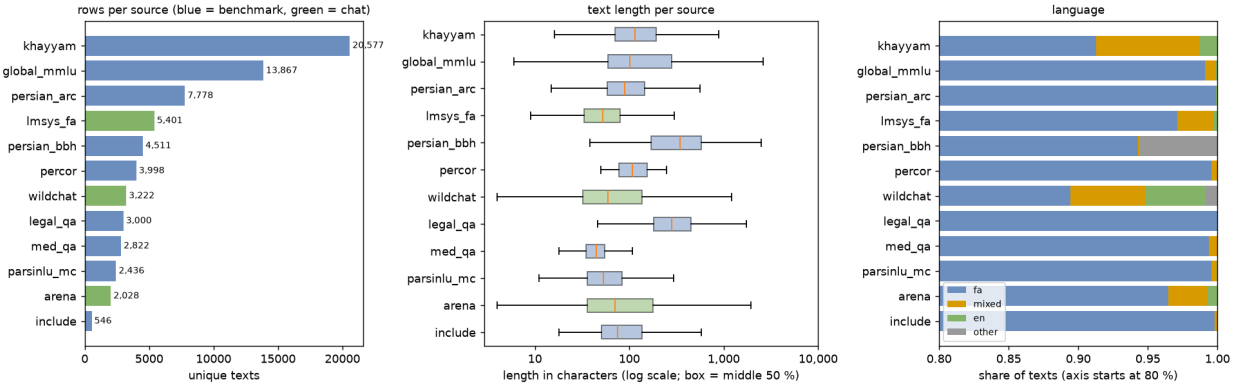


Figure 1: Data overview. Left: texts per source (blue benchmark, green chat). Middle: length on a log scale (box = middle 50 %). Right: language mix (axis starts at 80 %). Khayyam's "mixed" share comes from formulas; WildChat contains the most English.

3.2 Test sets

From these sources we build five test sets. Each one mirrors a situation from the paper or from our project:

Set	Size	Role
mix	28,802	nine benchmarks, at most 4,000 each; mirrors **EmbedLLM** (a mix of benchmarks)
Khayyam	20,577	native Persian school exams; 38 categories, 95 courses, 5 difficulty levels, examinee answer rates
Global-MMLU	13,867	machine-translated MMLU; 57 subjects in 6 categories
one topic	1,672	Khayyam's largest category (upper-secondary Persian literature); mirrors **Headlines**
chat	10,651	lmsys (5,401), WildChat (3,222), Arena (2,028); no labels; closest to Metis traffic

4 Method

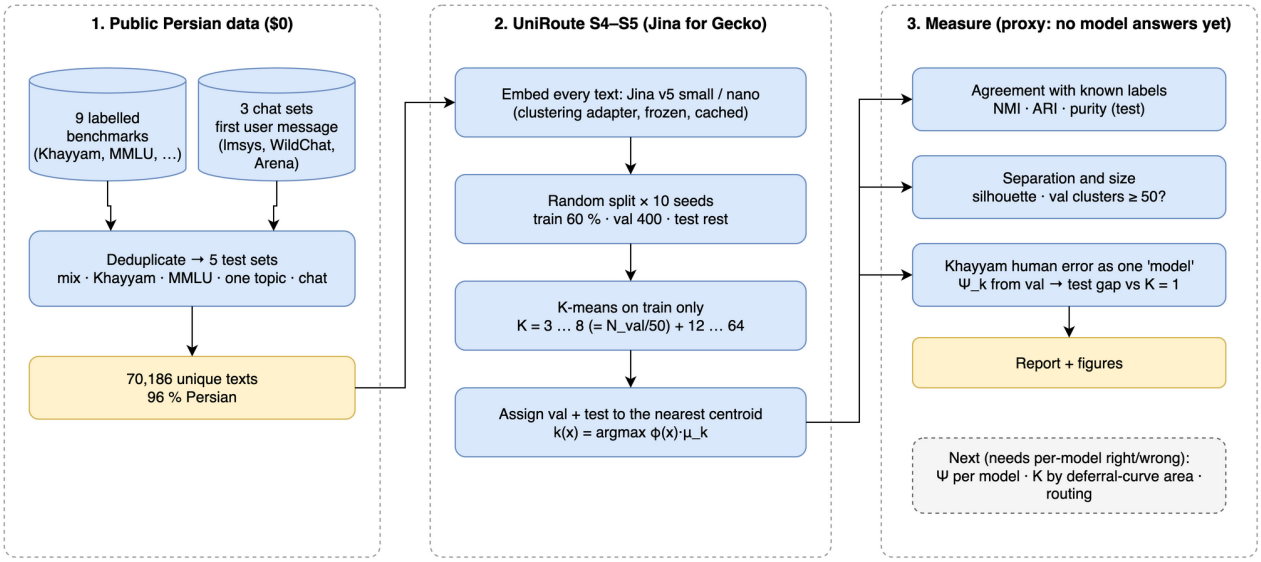


Figure 2: Pipeline of this study. Blue: free, offline steps; yellow: totals and outputs; grey: what still needs per-model labels.

4.1 Protocol (as in the paper)

- **Embedding.** Every unique text is embedded once with each embedder and cached by text hash. Inputs are capped at 2,048 tokens for this test; this affects 111 (small) / 71 (nano) chat messages.
- **Split.** For each set and each of R random seeds: train = 60 %, validation = 400 prompts (the project's N_{val}), test = the rest. The paper uses 400 repeats; we use $R = 10$ for $K \leq 8$ and $R = 3$ for larger K.
- **Clustering.** K-means (k-means++, 10 restarts, seed recorded) is fitted on the train part only. Validation and test prompts are assigned with Eq. (1).
- K **grid.** The paper's range $3 \ldots N_{\text{val}}/50 = 3 \ldots 8$, plus $K \in \{12, 16, 24, 32, 48, 64\}$ and $K =$ number of known classes, as references.
- **Statistics.** We report the mean over splits with a 95 % t-interval.

4.2 Measures

The paper chooses K by routing quality, which needs per-model labels. We use measures that do not need them.

NMI Normalised mutual information between test clusters and a known label: 0 = unrelated, 1 = identical. We also report the NMI with shuffled labels as the chance level.

ARI, purity Adjusted Rand index, and the share of test prompts that carry their cluster's most common label.

Silhouette Cosine silhouette on up to 3,000 test prompts: how much closer a prompt is to its own cluster than to the next one. It needs no labels.

Cluster size The number of validation clusters with fewer than 50 prompts. The spec (S5) requires $|C_k| \geq 50$ for every cluster.

Composition For one split at $K = 8$, the label mix of every cluster, and for every label the share of its prompts that land in its main cluster ("loyalty").

Topic anchors For unlabelled chat, we wrote 13 short topic descriptions in Persian and English (e.g. programming, law, poetry, small talk) and embedded them with the same model. Each prompt gets its nearest topic, and we report the topic mix per cluster. We validate the method on Khayyam, where subjects are known.

4.3 A routing proxy: human difficulty

Khayyam records, for 16,777 questions, the share of examinees who chose each option. We treat "the examinees" as one model with error $e(x) = 1-$(share choosing the key); blank answers count as wrong. In the spirit of Eq. (2), $\Psi_k = \overline{e}$ over the validation prompts of cluster k. On test prompts we measure the gap that Proposition 2 relates to routing loss, relative to one average for all prompts ($K = 1$):

$$\text{gap shrink}(K) = 1 - \frac{\mathbb{E}_{\text{test}}\,|e(x) - \Psi_{k(x)}|}{\mathbb{E}_{\text{test}}\,|e(x) - \overline{e}|}. \tag{4}$$

We compute Ψ twice. Once from the 400 validation prompts, which is the paper's setting. Once from the large train part, which is free of sampling noise. The difference between the two isolates the effect of the small validation set.

4.4 Environment

Apple M1 Pro (16 GB), PyTorch MPS, scikit-learn K-means, UMAP for the maps. A full run took 13 (nano) and 21 (small) minutes including new embeddings. Code: `scripts/s4_persian_cluster.py` and companions (Appendix C).

5 Results

5.1 Do clusters recover known topics? (RQ1, RQ2)

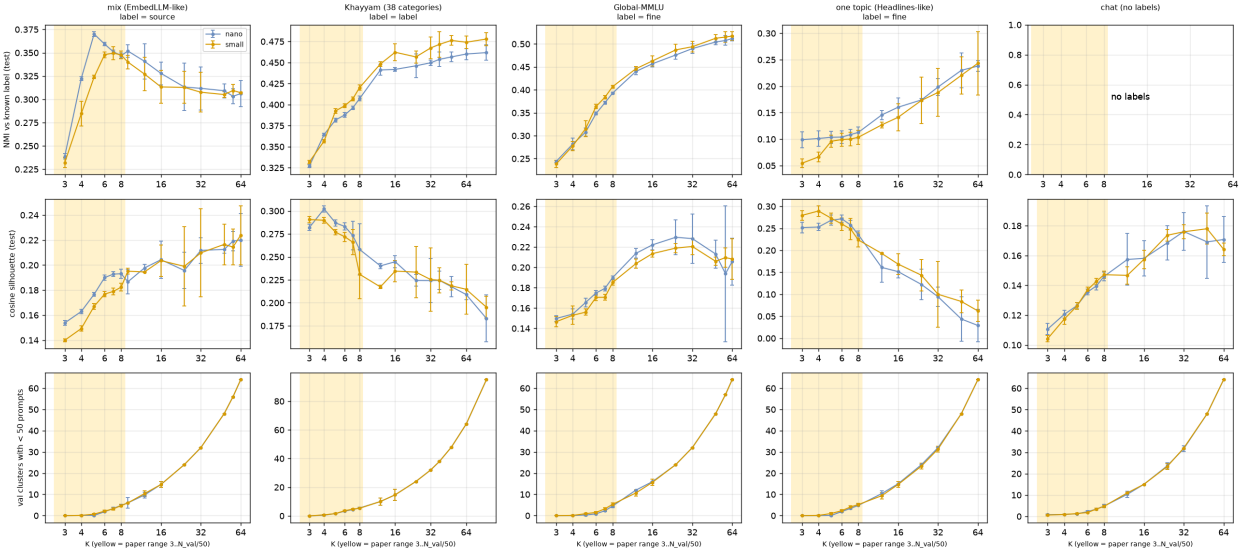


Figure 3: Scores against K for both embedders (mean and 95 % CI over splits). Top: NMI against the known label. Middle: cosine silhouette. Bottom: validation clusters with fewer than 50 prompts. Yellow band = the paper's range $K = 3 \ldots 8$ for $N_{\text{val}} = 400$.

Table 2: Paper range $K = 3 \ldots 8$: NMI ($\pm$ 95 % CI over 10 splits), silhouette, and number of validation clusters with fewer than 50 prompts. Labels: mix = source, Khayyam = 38 categories, Global-MMLU = 57 subjects, one topic = 6 courses.

		small			nano		
Set	K	NMI	Sil.	< 50	NMI	Sil.	< 50
mix	3	0.232 ± 0.005	0.140	0.0	0.239 ± 0.003	0.154	0.0
	4	0.285 ± 0.013	0.149	0.1	0.322 ± 0.002	0.163	0.1
	5	0.324 ± 0.002	0.167	0.7	0.370 ± 0.003	0.177	0.1
	6	0.348 ± 0.003	0.177	2.1	0.359 ± 0.002	0.190	1.8
	7	0.350 ± 0.007	0.179	3.2	0.351 ± 0.002	0.193	3.4
	8	0.348 ± 0.004	0.182	4.7	0.347 ± 0.003	0.193	4.8
Khayyam	3	0.332 ± 0.002	0.291	0.0	0.327 ± 0.002	0.282	0.0
	4	0.357 ± 0.002	0.290	0.7	0.364 ± 0.002	0.302	0.5
	5	0.392 ± 0.003	0.277	1.6	0.382 ± 0.002	0.287	1.6
	6	0.399 ± 0.002	0.272	3.6	0.388 ± 0.003	0.283	3.3
	7	0.407 ± 0.002	0.266	4.7	0.396 ± 0.002	0.273	4.4
	8	0.420 ± 0.003	0.231	5.3	0.407 ± 0.003	0.258	5.5
Global-MMLU	3	0.238 ± 0.008	0.147	0.0	0.244 ± 0.003	0.150	0.0
	4	0.278 ± 0.010	0.153	0.1	0.282 ± 0.013	0.154	0.0
	5	0.315 ± 0.017	0.156	0.9	0.308 ± 0.009	0.165	0.3
	6	0.364 ± 0.005	0.171	1.5	0.348 ± 0.003	0.175	0.8
	7	0.384 ± 0.004	0.171	3.2	0.372 ± 0.003	0.179	2.3
	8	0.407 ± 0.003	0.185	5.3	0.393 ± 0.002	0.190	4.5
one topic	3	0.054 ± 0.008	0.279	0.0	0.099 ± 0.015	0.252	0.0
	4	0.066 ± 0.009	0.289	0.1	0.101 ± 0.015	0.253	0.0
	5	0.096 ± 0.012	0.273	1.0	0.104 ± 0.011	0.269	0.1
	6	0.099 ± 0.012	0.260	2.3	0.104 ± 0.012	0.272	1.9
	7	0.100 ± 0.013	0.248	4.1	0.109 ± 0.010	0.258	3.4
	8	0.104 ± 0.013	0.224	5.2	0.113 ± 0.011	0.236	4.8
chat	3	–	0.104	0.6	–	0.111	0.8
	4	–	0.117	0.9	–	0.121	0.9
	5	–	0.126	1.3	–	0.126	1.2
	6	–	0.137	1.8	–	0.136	2.2
	7	–	0.143	3.4	–	0.140	3.4
	8	–	0.147	4.7	–	0.146	4.8

- **Benchmark mix.** NMI against the nine sources rises from 0.23 ($K = 3$) to 0.35–0.37 at $K = 5$–8, then falls slightly. Most sources form their own clusters (§5.4).
- **Subject-rich sets.** On Khayyam (38 categories) and Global-MMLU (57 subjects), NMI keeps rising with K (Khayyam 0.33 $\rightarrow$ 0.42 at $K = 8 \rightarrow$ 0.47 at $K = 38$; MMLU 0.24 $\rightarrow$ 0.41 $\rightarrow$ 0.52 at $K = 64$). These sets have more real subjects than eight clusters can hold.
- **One topic.** NMI stays near 0.10 in the paper range, and the silhouette falls quickly with K: there is little structure to find. This reproduces the paper's EmbedLLM-versus-Headlines contrast (**RQ2**).
- **Chat.** Silhouette 0.10–0.15 in the paper range, against 0.15–0.30 for the benchmarks. Chat clusters exist but are softer.
- **Cluster sizes.** With 400 validation prompts, $K = 8$ leaves on average about five of eight validation clusters below 50 prompts. Already at $K = 5$, one cluster often falls below 50.

5.2 small vs. nano vs. adapters (RQ3)

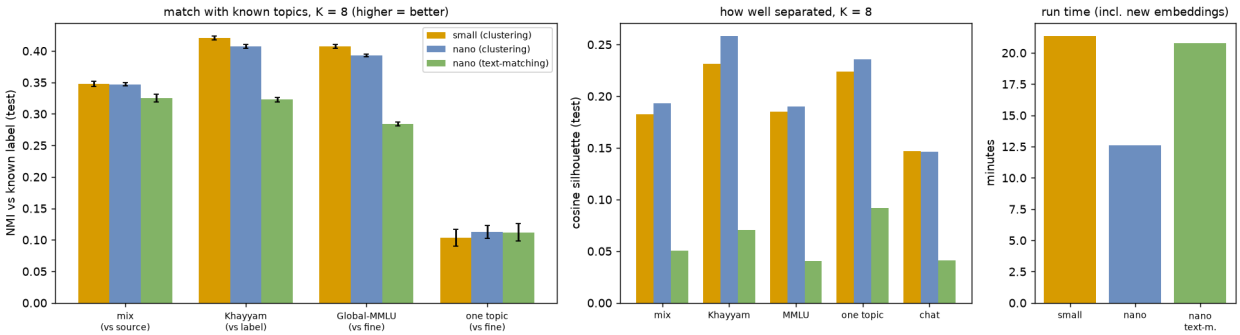


Figure 4: At $K = 8$: NMI against known labels, silhouette, and run time for small, nano and nano with the `text-matching` adapter.

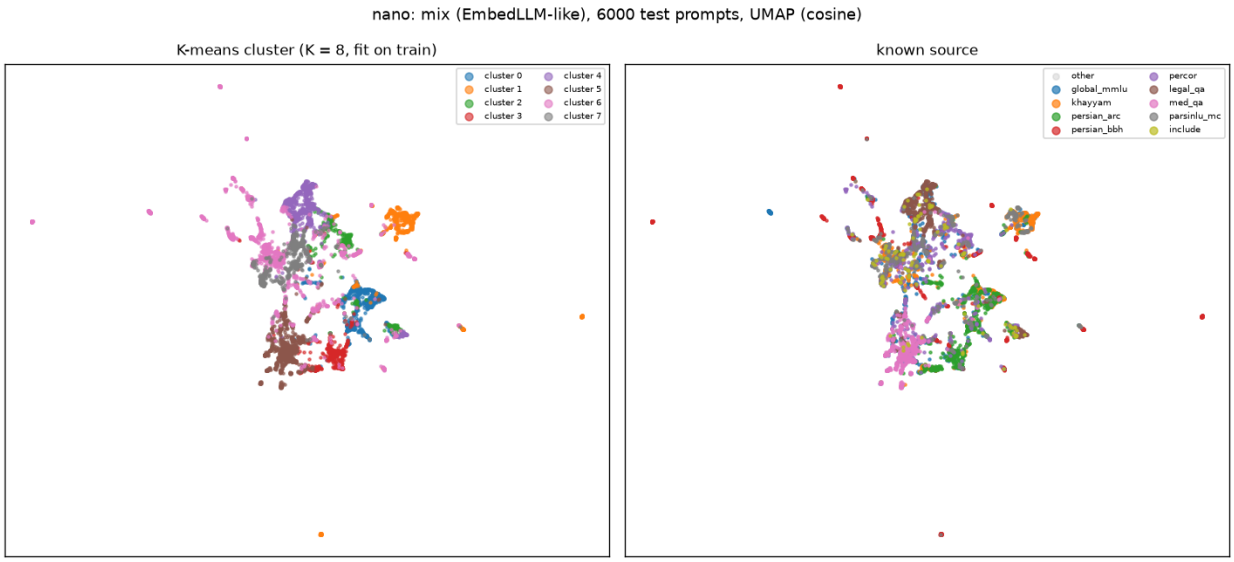
Table 3: Embedder comparison at $K = 8$ (NMI ± 95 % CI; chat: silhouette).

$K = 8$	small	nano	nano (text-m.)
mix (source)	0.348 ± 0.004	0.347 ± 0.003	0.325 ± 0.006
Khayyam (label)	0.420 ± 0.003	0.407 ± 0.003	0.323 ± 0.004
Global-MMLU (fine)	0.407 ± 0.003	0.393 ± 0.002	0.284 ± 0.003
one topic (fine)	0.104 ± 0.013	0.113 ± 0.011	0.112 ± 0.014
chat (no labels)	sil. 0.147	sil. 0.146	sil. 0.041
Khayyam difficulty gap shrink	5.4% ± 0.6%	5.4% ± 0.4%	7.7% ± 0.5%
Run time (min)	21	13	21

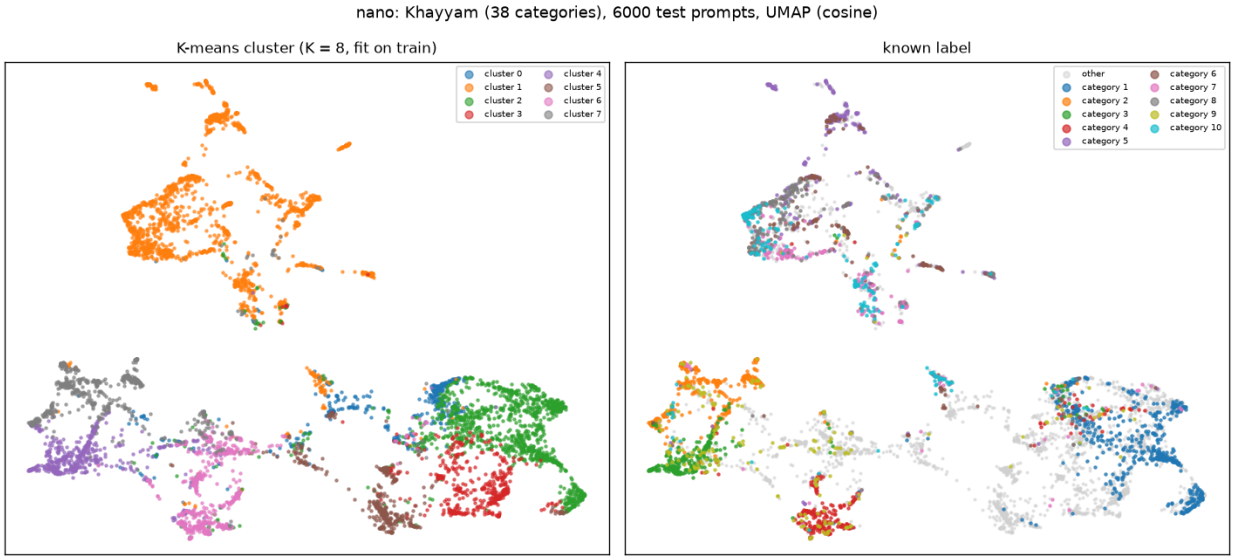
small beats *nano* by 0.013–0.014 NMI on the two subject-rich sets. *nano* wins on the mix at $K = 4$–5 (0.370 vs. 0.324 at $K = 5$) and on the single topic, and it is better separated on every labelled set. Both shrink the Khayyam difficulty gap equally. For clustering alone, the two are interchangeable, and nano is faster and the only one that meets the latency budget. The `text-matching` adapter clusters clearly worse: NMI 0.08–0.12 lower, silhouette near 0.05, and lopsided cluster sizes. Unexpectedly, its difficulty gain is larger (7.7 %); we return to this in §6.

5.3 Cluster maps

Figure 5 projects 6,000 test prompts per set to two dimensions with UMAP. On the left, dots are coloured by K-means cluster ($K = 8$); on the right, by known label. Where the coloured islands on both sides coincide, the clusters follow the labels.

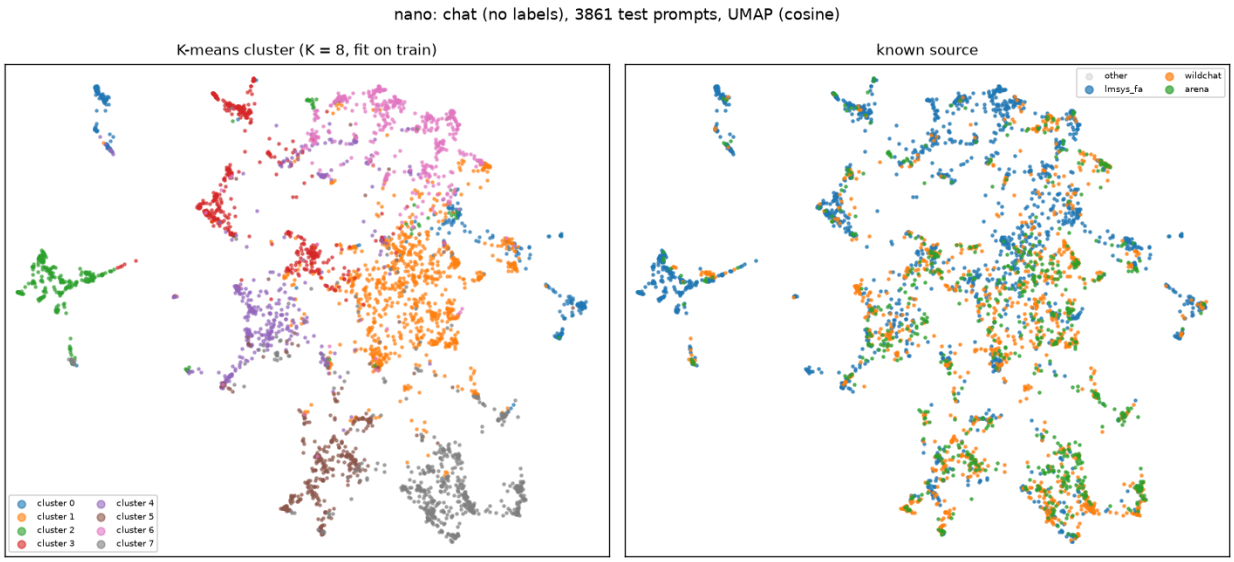


(a) mix: right = source



(b) Khayyam: right = ten largest categories (numbered as in Fig. 7)

Figure 5: UMAP maps (nano, $K = 8$, one split, test prompts).



(c) chat: right = source dataset

Figure 5: (continued) On chat, the three sources are largely mixed; clusters follow what is asked, not where the data came from.

5.4 Are the clusters meaningful? (RQ4)

Better than chance. At $K = 8$ (nano, one split), NMI against the known label is far above the shuffled baseline: Khayyam 0.405 vs. 0.005, Global-MMLU subjects 0.392 vs. 0.015, mix sources 0.350 vs. 0.001. Khayyam's difficulty *level* reaches only 0.009, so clusters group by subject, not by difficulty.

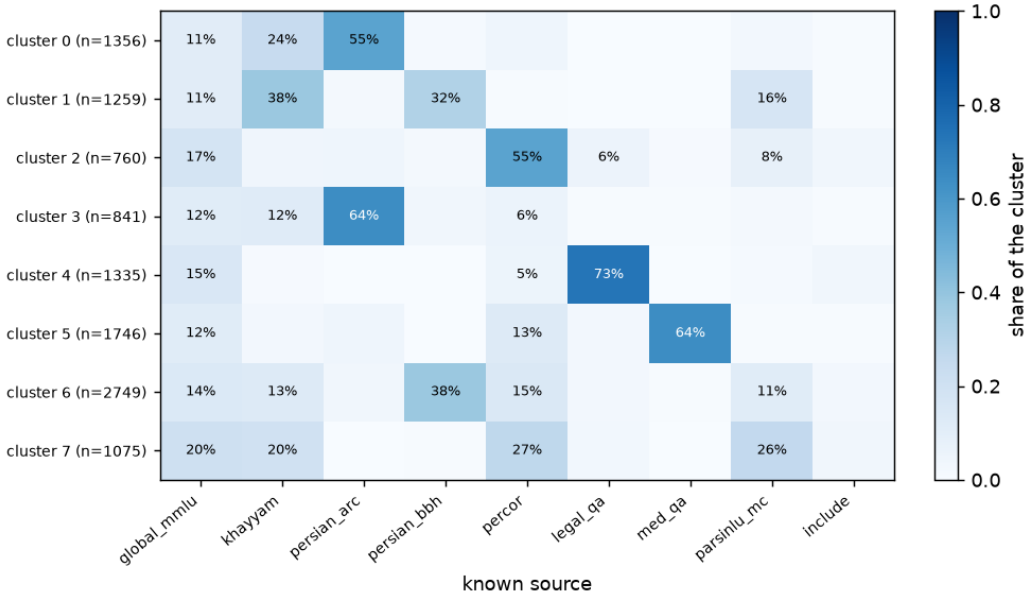


Figure 6: mix: share of each source inside each cluster (rows sum to 100 %).

Table 4: mix: largest sources per cluster.

#	n	Largest sources (share of cluster)
0	1,356	persian_arc 55%; khayyam 24%; global_mmlu 11%; percor 5%
1	1,259	khayyam 38%; persian_bbh 32%; parsinlu_mc 16%; global_mmlu 11%
2	760	percor 55%; global_mmlu 17%; parsinlu_mc 8%; legal_qa 6%
3	841	persian_arc 64%; khayyam 12%; global_mmlu 12%; percor 6%
4	1,335	legal_qa 73%; global_mmlu 15%; percor 5%; include 3%
5	1,746	med_qa 64%; percor 13%; global_mmlu 12%; persian_arc 5%
6	2,749	persian_bbh 38%; percor 15%; global_mmlu 14%; khayyam 13%
7	1,075	percor 27%; parsinlu_mc 26%; global_mmlu 20%; khayyam 20%

Benchmark mix. Legal QA (73 % of cluster 4), medical QA (64 % of cluster 5), PerCoR news and commonsense (55 % of cluster 2) and Persian ARC science (clusters 0 and 3) each dominate a cluster. Reasoning puzzles from BBH share clusters 1 and 6 with Khayyam's mathematics and logic questions. Clusters follow *what is asked*, not *which dataset* the question came from. Global-MMLU spreads over all clusters because it covers 57 subjects.

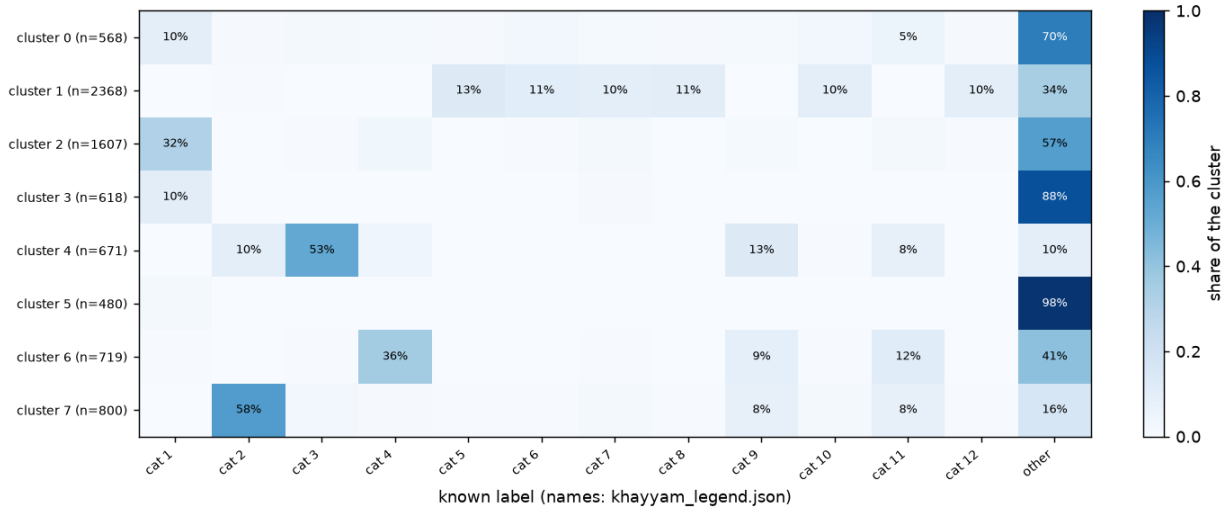


Figure 7: Khayyam: share of the twelve largest categories inside each cluster. cat 1 Persian literature (upper sec.), 2 physics, 3 chemistry, 4 biology, 5 mathematics (upper sec.), 6 mathematics & statistics, 7 mathematical & logical IQ (upper prim.), 8 mathematics (lower sec.), 9 science (lower sec.), 10 mathematics (upper prim.), 11 science (upper prim.), 12 geometry; "other" = the 26 smaller categories.

Table 5: Khayyam clusters ($K = 8$, nano, test prompts): name, size, largest categories, and mean examinee error.

#	Name	n	Largest categories (share of cluster)	Error
0	humanities mix	568	Psychology (upper sec.) 11%; Economics (upper sec.) 10%; Persian literature (upper sec.) 10%	0.54
1	mathematics	2,368	Mathematics (upper sec.) 13%; Calculus (upper sec.) 12%; Mathematics & statistics (upper sec.) 11%	0.71
2	Persian literature	1,607	Persian literature (upper sec.) 32%; Persian literature (lower sec.) 10%; Persian literature (upper prim.) 10%	0.56
3	religion & philosophy	618	Philosophy (upper sec.) 18%; Islamic education (upper prim.) 11%; Persian literature (upper sec.) 10%	0.56
4	chemistry	671	Chemistry (upper sec.) 53%; Science (lower sec.) 13%; Physics (upper sec.) 10%	0.66
5	history & society	480	History (upper sec.) 38%; Sociology (upper sec.) 20%; Social studies (lower sec.) 14%	0.58
6	biology & science	719	Biology (upper sec.) 36%; Science (upper prim.) 12%; Geography (upper sec.) 10%	0.57
7	physics	800	Physics (upper sec.) 58%; Science (lower sec.) 8%; Science (upper prim.) 8%	0.69

Khayyam. Eight clusters must hold 38 categories, so related subjects share. Every cluster can still be named (Table 5), and loyalty is high (Appendix A): calculus 99.6 %, upper-secondary mathematics 95 %, discrete mathematics 95 %, chemistry 87 %, physics 83 %, Persian literature 79–87 % at all four school levels, religion 83–100 %. The errors are sensible. All mathematics merges into one cluster regardless of grade. Lower-grade "general science" and social studies split (30–37 % in their main cluster), which matches their mixed content. Clusters also **differ in difficulty**: examinees miss 71 % of mathematics and 69 % of physics questions, but only 54–58 % in the humanities clusters. That is the kind of signal Eq. (3) uses.

Table 6: Global-MMLU clusters ($K = 8$, nano).

#	n	Largest subjects	Largest categories
0	684	high school macroeconomics 19%; professional accounting 16%; high school microeconomics 12%	Social Sciences 47%; Business 30%
1	542	conceptual physics 13%; high school chemistry 12%; miscellaneous 9%	STEM 72%; Other 9%
2	806	professional medicine 12%; nutrition 12%; clinical knowledge 10%	Medical 51%; STEM 23%
3	573	high school world history 13%; high school us history 13%; prehistory 9%	Humanities 60%; Social Sciences 28%
4	545	elementary mathematics 21%; high school mathematics 15%; high school statistics 7%	STEM 78%; Social Sciences 12%
5	556	professional law 85%; international law 4%; high school government and politics 3%	Humanities 92%; Social Sciences 6%
6	639	professional psychology 13%; high school psychology 13%; miscellaneous 8%	Social Sciences 42%; Humanities 26%
7	802	moral scenarios 42%; philosophy 8%; miscellaneous 7%	Humanities 71%; Social Sciences 17%

Global-MMLU. Each cluster has a clear theme: economics and accounting, physical sciences, medicine, history, mathematics, law (85 % professional law), psychology and philosophy, and moral scenarios. Medical and business questions each land 77 % in one cluster. The "humanities" category splits into history, law and ethics, which really are different subjects.

Chat (no labels). The topic-anchor method agrees with Khayyam's known subject on 71.9 % of 6,574 mapped questions, against 18.7 % by chance. It is rough but informative. Figure 8 and Table 7 apply it to chat.

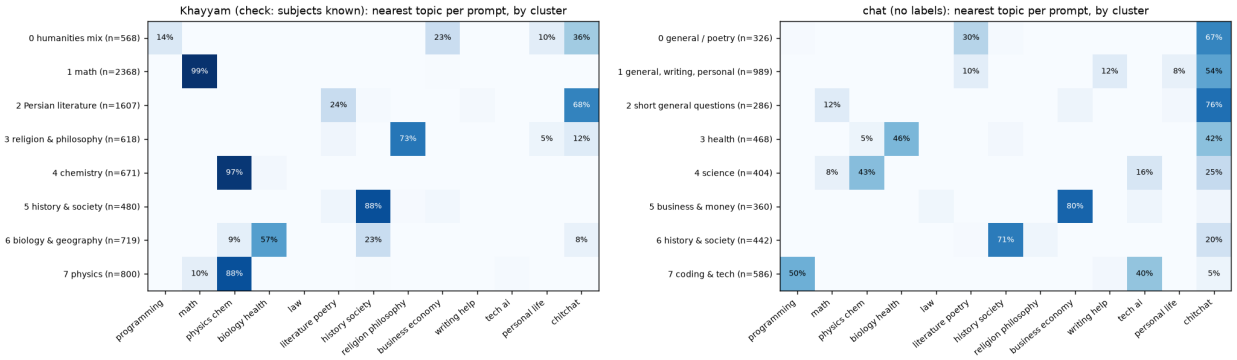


Figure 8: Nearest topic per prompt, by cluster (top four topics per cluster shown). Left: Khayyam, where subjects are known (validation of the method). Right: chat.

Table 7: Chat clusters ($K = 8$, nano). Nearest topic (top three); median length; share of Latin letters; share with code or math markers; largest source.

#	Name	n	Nearest topic	Chars	Latin	Code/math	Main source
0	general / poetry	326	chitchat 67%; literature poetry 30%; history society 2%	58	3%	0%	lmsys_fa 82%
1	general, writing, personal	989	chitchat 54%; writing help 12%; literature poetry 10%	48	4%	1%	wildchat 40%
2	short general questions	286	chitchat 76%; math 12%; business economy 5%	55	1%	1%	lmsys_fa 84%
3	health	468	biology health 46%; chitchat 42%; physics chem 5%	52	2%	1%	lmsys_fa 69%
4	science	404	physics chem 43%; chitchat 25%; tech ai 16%	62	3%	2%	lmsys_fa 55%
5	business & money	360	business economy 80%; tech ai 4%; chitchat 4%	64	5%	5%	wildchat 47%
6	history & society	442	history society 71%; chitchat 20%; religion philosophy 4%	50	2%	1%	lmsys_fa 76%
7	coding & tech	586	programming 50%; tech ai 40%; chitchat 5%	72	10%	6%	wildchat 44%

Five of eight chat clusters carry a clear topic. Coding and technology (cluster 7) is also the cluster with the most Latin script and code markers, which independently confirms the label. The others are business (80 %), history and society (71 %), health and science. The remaining three are general, which here also means "no anchor fits". Clusters 0, 2 and 6 come mostly from lmsys_fa, a translated set, so part of their grouping may reflect translation style.

5.5 Would these clusters help routing? (RQ5)

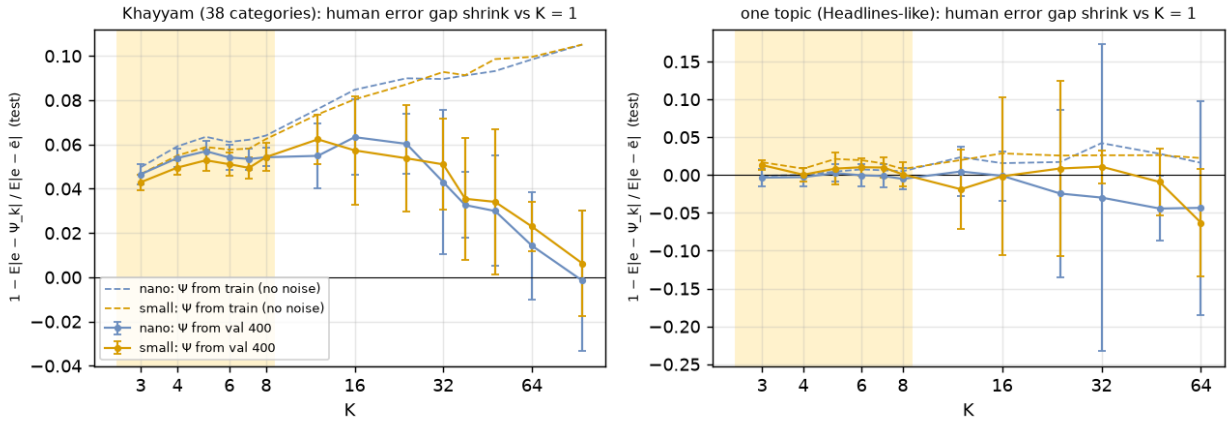


Figure 9: Gap shrink against $K = 1$ on test prompts. Solid: Ψ from 400 validation prompts (paper setting); dashed: Ψ from the large train part.

Table 8: Gap shrink in % (validation rows: ± 95 % CI).

Set	Model	Profile	$K = 3$	$K = 5$	$K = 8$	$K = 16$	$K = 32$	$K = 64$
Khayyam	small	Ψ from val (400)	4.3±0.3	5.3±0.5	5.4±0.6	5.7±2.4	5.1±2.1	2.3±1.1
Khayyam	small	Ψ from train	4.7	5.9	6.2	8.0	9.3	10.0
Khayyam	nano	Ψ from val (400)	4.7±0.5	5.7±0.4	5.4±0.4	6.3±1.7	4.3±3.3	1.4±2.4
Khayyam	nano	Ψ from train	5.0	6.3	6.4	8.5	9.0	9.8
one topic	small	Ψ from val (400)	1.3±0.6	0.8±2.1	0.1±1.6	-0.2±10.4	1.1±2.2	-6.3±7.1
one topic	small	Ψ from train	1.7	2.1	0.8	2.9	2.6	2.2
one topic	nano	Ψ from val (400)	-0.3±1.1	0.3±1.1	-0.5±1.4	-0.1±3.3	-3.0±20.2	-4.3±14.1
one topic	nano	Ψ from train	-0.1	0.5	0.5	1.6	4.2	1.6

On Khayyam, clusters shrink the error gap by 4–6 % across the paper range. On the single topic the shrink is zero within the confidence interval. A router fed with these clusters would learn something about difficulty on a subject mix and nothing on a single topic, which is again the paper's contrast. When the profiles come from the large train part, the gain keeps growing to about 10 % at $K = 64$. With only 400 validation prompts it peaks at $K \approx 8$–16 and then falls, because clusters with a handful of prompts give noisy profiles. **The validation size, not the embedder, limits the useful K.** This is the reason for the paper's cap $K \leq N_{\text{val}}/50$.

6 Discussion

Answers to the research questions.

RQ1 Yes. Jina v5 plus K-means recovers known Persian topics far above chance. Agreement rises with K up to the number of real subjects.

RQ2 Yes. A benchmark mix clusters well, a single topic hardly at all, and real chat lies in between.

RQ3 *nano* with the `clustering` adapter. It is within 0.015 NMI of *small*, better separated, faster, and the only variant that meets the latency budget. The final choice stays with WP-08 on real data (D-052).

RQ4 Yes, on labelled data. Clusters are nameable and loyal, and their errors are sensible merges of neighbouring subjects. On chat, five of eight clusters are topical. A human check of sample prompts is still needed to confirm the chat names.

RQ5 Likely, on mixed traffic. Clusters separate easy from hard subjects, but the gain is modest (about 5 %) and capped by the 400-prompt validation set.

Implications for the project.

- K **and the cluster-size rule (Q-024, decided).** The spec requires *every* validation cluster to hold at least 50 prompts; the paper only caps $K \leq N_{\text{val}}/50$, i.e. 50 *on average*. With 400 validation prompts the strict rule leaves $K = 3$–4. The paper's rule allows up to 8, where our gains peak. Decision (D-054): follow the paper, with no per-cluster minimum and $\alpha = 0$ (Eq. 12). A larger validation set would raise the ceiling, at a cost.
- **Real traffic will look like the chat set,** not like the benchmarks: softer clusters and a large "general" share. Expect smaller per-cluster differences between models than on Khayyam.
- **Adapters.** The `clustering` adapter is the right default for routing. The larger difficulty gain of `text-matching`, despite worse topic clusters, suggests it may capture form (length, formulas, question type) that relates to difficulty. This should be re-checked when real per-model labels exist.

Comparison with the paper. Our protocol matches the paper's in embedding, train-only K-means, nearest-centroid assignment, the validation size and the K range. It differs in three ways. The embedder is Jina instead of Gecko. We use 10 splits instead of 400. And we use label-based and human-difficulty proxies instead of the deferral-curve area, which needs per-model correctness.

7 Threats to validity

- **No model answers.** We measure cluster quality and a human-difficulty proxy, not routing quality. Human difficulty is one "model"; LLMs may find different questions hard.
- **Domain shift.** Exam questions differ from chat. The chat set is the closest stand-in, but lmsys_fa is translated, and the three chat sources may not match Metis users.
- **Labels as ground truth.** Source and subject labels are imperfect topic labels. Some benchmark questions cross subjects, and Global-MMLU is machine-translated.
- **Topic anchors** are hand-written and approximate (72 % on Khayyam). The "small talk" anchor also catches anything that fits no topic.
- **Statistics.** Ten splits ($K \leq 8$) and three ($K > 8$) give narrow intervals for NMI, but wider ones for large K. Maps and compositions use one split.
- **Truncation** at 2,048 tokens affects about 1 % of chat messages in this test only.

8 Conclusion and next steps

On Persian data, the embedding and clustering stage of UniRoute behaves as the paper describes. With Jina v5 nano it produces meaningful, nameable clusters on mixed content and weak ones on a single topic. Its benefit is limited more by the size of the labelled validation set than by the embedder. Next steps: (1) the author checks the chat cluster names on the local sample pages; (2) optionally, once the router core (WP-05) exists, run the full UniRoute pipeline on public data with per-model labels, either EmbedLLM to reproduce the paper or Khayyam answered by local open models, to measure routing quality directly.

References

[1] W. Jitkrittum, H. Narasimhan, A. S. Rawat, J. Juneja, C. Wang, Z. Wang, A. Go, C.-Y. Lee, P. Shenoy, R. Panigrahy, A. K. Menon, S. Kumar. *Universal Model Routing for Efficient LLM Inference.* ICLR 2026.

[2] Khayyam Challenge (PersianMMLU), `raia-center/khayyam-challenge`, arXiv:2404.06644.

[3] Global-MMLU, `CohereLabs/Global-MMLU`, arXiv:2412.03304.

[4] Jina AI, *jina-embeddings-v5-text* (small, nano), Hugging Face model cards.

A Khayyam: where each category lands

Share of each category's test questions that fall in its main cluster (nano, $K = 8$, one split).

Khayyam category	n (test)	Main cluster	Share there
Islamic education (lower prim.)	17	3 (religion & philosophy)	100%
Calculus (upper sec.)	285	1 (mathematics)	100%
Religion & life (upper sec.)	55	3 (religion & philosophy)	96%
Mathematics (upper sec.)	329	1 (mathematics)	95%
Discrete mathematics (upper sec.)	128	1 (mathematics)	94%
Mathematics (lower sec.)	265	1 (mathematics)	94%
Geometry (upper sec.)	258	1 (mathematics)	93%
Statistics & probability (upper sec.)	134	1 (mathematics)	91%
Mathematics & statistics (upper sec.)	287	1 (mathematics)	90%
Islamic education (lower sec.)	72	3 (religion & philosophy)	89%
Chemistry (upper sec.)	406	4 (chemistry)	87%
Persian literature (lower prim.)	134	2 (Persian literature)	87%
Mathematics (upper prim.)	280	1 (mathematics)	85%
Persian literature (upper prim.)	193	2 (Persian literature)	84%
Islamic education (upper prim.)	84	3 (religion & philosophy)	83%
Physics (upper sec.)	555	7 (physics)	83%
Mathematics (lower prim.)	192	1 (mathematics)	82%
History (upper sec.)	228	5 (history & society)	81%
Verbal & literary IQ (upper prim.)	59	2 (Persian literature)	80%
Persian literature (upper sec.)	645	2 (Persian literature)	79%
Persian literature (lower sec.)	212	2 (Persian literature)	79%
Mathematical & logical IQ (upper prim.)	288	1 (mathematics)	78%
Philosophy (upper sec.)	153	3 (religion & philosophy)	74%
Economics (upper sec.)	78	0 (humanities mix)	72%
Biology (upper sec.)	365	6 (biology & science)	70%
Logic (upper sec.)	74	2 (Persian literature)	54%
Sociology (upper sec.)	184	5 (history & society)	51%
Psychology (upper sec.)	127	0 (humanities mix)	49%
Speed & accuracy (upper prim.)	93	1 (mathematics)	44%
Analytical aptitude (lower sec.)	156	2 (Persian literature)	44%
Social studies (lower prim.)	62	2 (Persian literature)	42%
Social studies (lower sec.)	180	5 (history & society)	37%
Geography (upper sec.)	196	6 (biology & science)	36%
Geology (upper sec.)	157	6 (biology & science)	35%
Science (lower sec.)	257	4 (chemistry)	35%
Science (lower prim.)	194	6 (biology & science)	34%
Science (upper prim.)	267	6 (biology & science)	31%
Social studies (upper prim.)	182	5 (history & society)	30%

B Full results per K

NMI, ARI and purity against the main label of each set; silhouette; smallest validation cluster; number of validation clusters below 50; largest test-cluster share. Mean over splits (NMI ± 95 % CI).

Jina v5 small

K	splits	NMI	ARI	Purity	Sil.	val min	val < 50	max share
mix ($n = 28,802$; label: source)								
3	10	0.232 ± 0.005	0.129	0.309	0.140	83	0.0	0.50
4	10	0.285 ± 0.013	0.175	0.383	0.149	60	0.1	0.43
5	10	0.324 ± 0.002	0.223	0.444	0.167	43	0.7	0.33
6	10	0.348 ± 0.003	0.249	0.486	0.177	35	2.1	0.30
7	10	0.350 ± 0.007	0.262	0.505	0.179	32	3.2	0.25
8	10	0.348 ± 0.004	0.258	0.514	0.182	28	4.7	0.22
9	3	0.340 ± 0.007	0.236	0.499	0.195	25	6.0	0.21
12	3	0.327 ± 0.018	0.230	0.519	0.195	19	10.3	0.16
16	3	0.313 ± 0.018	0.200	0.518	0.204	10	14.7	0.13
24	3	0.313 ± 0.017	0.168	0.527	0.199	4	24.0	0.11
32	3	0.308 ± 0.021	0.148	0.537	0.210	2	32.0	0.10
48	3	0.305 ± 0.003	0.101	0.552	0.217	0	48.0	0.05
56	3	0.309 ± 0.006	0.094	0.571	0.214	0	56.0	0.04
64	3	0.307 ± 0.001	0.084	0.574	0.224	0	64.0	0.04
Khayyam ($n = 20,577$; label: label)								
3	10	0.332 ± 0.002	0.099	0.182	0.291	104	0.0	0.42
4	10	0.357 ± 0.002	0.120	0.212	0.290	44	0.7	0.38
5	10	0.392 ± 0.003	0.155	0.241	0.277	44	1.6	0.31
6	10	0.399 ± 0.002	0.159	0.271	0.272	32	3.6	0.30
7	10	0.407 ± 0.002	0.170	0.279	0.266	30	4.7	0.30
8	10	0.420 ± 0.003	0.193	0.295	0.231	27	5.3	0.25
12	3	0.448 ± 0.003	0.230	0.349	0.217	13	10.0	0.16
16	3	0.462 ± 0.010	0.243	0.367	0.234	8	14.7	0.13
24	3	0.457 ± 0.007	0.226	0.382	0.233	2	24.0	0.10
32	3	0.467 ± 0.013	0.235	0.407	0.225	2	32.0	0.08
38	3	0.472 ± 0.015	0.227	0.423	0.225	1	38.0	0.08
48	3	0.476 ± 0.006	0.226	0.439	0.218	0	48.0	0.05
64	3	0.474 ± 0.007	0.201	0.447	0.215	0	64.0	0.05
95	3	0.478 ± 0.008	0.167	0.475	0.195	0	95.0	0.04
Global-MMLU ($n = 13,867$; label: fine)								
3	10	0.238 ± 0.008	0.067	0.151	0.147	93	0.0	0.46
4	10	0.278 ± 0.010	0.076	0.170	0.153	70	0.1	0.38
5	10	0.315 ± 0.017	0.111	0.195	0.156	44	0.9	0.33
6	10	0.364 ± 0.005	0.165	0.252	0.171	41	1.5	0.27
7	10	0.384 ± 0.004	0.187	0.260	0.171	35	3.2	0.21
8	10	0.407 ± 0.003	0.210	0.272	0.185	30	5.3	0.21
12	3	0.446 ± 0.005	0.229	0.329	0.204	15	10.7	0.16
16	3	0.463 ± 0.011	0.260	0.354	0.214	9	15.7	0.14
24	3	0.487 ± 0.012	0.295	0.401	0.219	4	24.0	0.11
32	3	0.494 ± 0.011	0.303	0.424	0.221	2	32.0	0.09
48	3	0.512 ± 0.008	0.316	0.463	0.206	1	48.0	0.08
57	3	0.515 ± 0.012	0.318	0.470	0.210	1	57.0	0.08
64	3	0.517 ± 0.010	0.309	0.480	0.208	0	64.0	0.08
one topic ($n = 1,672$; label: fine)								
3	10	0.054 ± 0.008	0.026	0.319	0.279	99	0.0	0.48

K	splits	NMI	ARI	Purity	Sil.	val min	val < 50	max share
4	10	0.066 ± 0.009	0.033	0.329	0.289	61	0.1	0.38
5	10	0.096 ± 0.012	0.049	0.366	0.273	41	1.0	0.36
6	10	0.099 ± 0.012	0.046	0.367	0.260	35	2.3	0.33
7	10	0.100 ± 0.013	0.046	0.372	0.248	25	4.1	0.31
8	10	0.104 ± 0.013	0.043	0.380	0.224	22	5.2	0.30
12	3	0.127 ± 0.006	0.040	0.408	0.193	7	9.3	0.19
16	3	0.141 ± 0.025	0.038	0.421	0.169	1	14.7	0.17
24	3	0.173 ± 0.044	0.038	0.461	0.143	1	23.3	0.13
32	3	0.188 ± 0.045	0.029	0.465	0.100	0	31.3	0.12
48	3	0.221 ± 0.035	0.025	0.493	0.084	0	48.0	0.09
64	3	0.243 ± 0.060	0.021	0.515	0.062	0	64.0	0.08
chat ($n = 10,651$; label: none)								
3	10	–	–	–	0.104	57	0.6	0.55
4	10	–	–	–	0.117	41	0.9	0.39
5	10	–	–	–	0.126	38	1.3	0.34
6	10	–	–	–	0.137	35	1.8	0.30
7	10	–	–	–	0.143	31	3.4	0.26
8	10	–	–	–	0.147	27	4.7	0.24
12	3	–	–	–	0.147	18	11.0	0.18
16	3	–	–	–	0.157	10	15.0	0.14
24	3	–	–	–	0.174	5	23.3	0.10
32	3	–	–	–	0.176	3	32.0	0.08
48	3	–	–	–	0.178	1	48.0	0.06
64	3	–	–	–	0.164	0	64.0	0.04

Jina v5 nano

K	splits	NMI	ARI	Purity	Sil.	val min	val < 50	max share
mix ($n = 28,802$; label: source)								
3	10	0.239 ± 0.003	0.140	0.303	0.154	75	0.0	0.40
4	10	0.322 ± 0.002	0.231	0.432	0.163	66	0.1	0.33
5	10	0.370 ± 0.003	0.290	0.502	0.177	55	0.1	0.28
6	10	0.359 ± 0.002	0.275	0.492	0.190	32	1.8	0.23
7	10	0.351 ± 0.002	0.258	0.496	0.193	26	3.4	0.22
8	10	0.347 ± 0.003	0.252	0.502	0.193	22	4.8	0.23
9	3	0.352 ± 0.007	0.266	0.524	0.187	23	6.0	0.20
12	3	0.341 ± 0.019	0.255	0.530	0.198	12	9.7	0.15
16	3	0.328 ± 0.012	0.242	0.525	0.204	4	14.7	0.14
24	3	0.313 ± 0.025	0.184	0.535	0.196	2	24.0	0.11
32	3	0.312 ± 0.023	0.157	0.541	0.212	1	32.0	0.10
48	3	0.309 ± 0.008	0.105	0.559	0.213	0	48.0	0.05
56	3	0.303 ± 0.008	0.095	0.556	0.219	0	56.0	0.05
64	3	0.306 ± 0.014	0.086	0.567	0.220	0	64.0	0.04
Khayyam ($n = 20,577$; label: label)								
3	10	0.327 ± 0.002	0.099	0.180	0.282	104	0.0	0.41
4	10	0.364 ± 0.002	0.122	0.224	0.302	49	0.5	0.38
5	10	0.382 ± 0.002	0.143	0.240	0.287	39	1.6	0.31
6	10	0.388 ± 0.003	0.147	0.271	0.283	31	3.3	0.30
7	10	0.396 ± 0.002	0.161	0.281	0.273	27	4.4	0.30
8	10	0.407 ± 0.003	0.178	0.291	0.258	22	5.5	0.28
12	3	0.441 ± 0.006	0.234	0.341	0.240	8	10.0	0.18
16	3	0.442 ± 0.002	0.237	0.357	0.245	6	14.7	0.13
24	3	0.446 ± 0.014	0.227	0.384	0.224	1	24.0	0.10
32	3	0.450 ± 0.003	0.211	0.403	0.224	0	32.0	0.09

K	splits	NMI	ARI	Purity	Sil.	val min	val < 50	max share
38	3	0.454 ± 0.009	0.215	0.409	0.224	1	38.0	0.07
48	3	0.457 ± 0.006	0.204	0.421	0.218	0	48.0	0.07
64	3	0.460 ± 0.007	0.196	0.436	0.209	0	64.0	0.05
95	3	0.462 ± 0.009	0.162	0.451	0.183	0	95.0	0.04
Global-MMLU ($n = 13,867$; label: fine)								
3	10	0.244 ± 0.003	0.076	0.153	0.150	94	0.0	0.41
4	10	0.282 ± 0.013	0.092	0.167	0.154	78	0.0	0.32
5	10	0.308 ± 0.009	0.108	0.178	0.165	55	0.3	0.25
6	10	0.348 ± 0.003	0.156	0.239	0.175	46	0.8	0.22
7	10	0.372 ± 0.003	0.179	0.256	0.179	42	2.3	0.19
8	10	0.393 ± 0.002	0.204	0.267	0.190	39	4.5	0.16
12	3	0.440 ± 0.007	0.266	0.323	0.214	18	12.0	0.12
16	3	0.457 ± 0.007	0.262	0.352	0.222	12	16.0	0.10
24	3	0.476 ± 0.010	0.282	0.384	0.230	2	24.0	0.09
32	3	0.490 ± 0.011	0.312	0.417	0.228	2	32.0	0.08
48	3	0.504 ± 0.007	0.315	0.452	0.213	0	48.0	0.07
57	3	0.508 ± 0.011	0.290	0.459	0.194	0	57.0	0.07
64	3	0.512 ± 0.005	0.308	0.471	0.206	0	64.0	0.07
one topic ($n = 1,672$; label: fine)								
3	10	0.099 ± 0.015	0.083	0.375	0.252	72	0.0	0.54
4	10	0.101 ± 0.015	0.069	0.377	0.253	66	0.0	0.35
5	10	0.104 ± 0.011	0.060	0.376	0.269	58	0.1	0.28
6	10	0.104 ± 0.012	0.061	0.378	0.272	38	1.9	0.28
7	10	0.109 ± 0.010	0.057	0.380	0.258	28	3.4	0.25
8	10	0.113 ± 0.011	0.054	0.393	0.236	20	4.8	0.24
12	3	0.146 ± 0.009	0.053	0.428	0.161	5	10.3	0.17
16	3	0.160 ± 0.018	0.048	0.440	0.151	4	15.0	0.15
24	3	0.174 ± 0.003	0.038	0.452	0.122	0	24.0	0.10
32	3	0.198 ± 0.016	0.033	0.468	0.094	0	32.0	0.09
48	3	0.230 ± 0.033	0.031	0.512	0.044	0	48.0	0.09
64	3	0.238 ± 0.010	0.020	0.519	0.029	0	64.0	0.08
chat ($n = 10,651$; label: none)								
3	10	–	–	–	0.111	47	0.8	0.57
4	10	–	–	–	0.121	35	0.9	0.41
5	10	–	–	–	0.126	31	1.2	0.34
6	10	–	–	–	0.136	31	2.2	0.31
7	10	–	–	–	0.140	28	3.4	0.28
8	10	–	–	–	0.146	26	4.8	0.26
12	3	–	–	–	0.157	17	10.3	0.18
16	3	–	–	–	0.158	8	15.0	0.17
24	3	–	–	–	0.168	4	23.7	0.11
32	3	–	–	–	0.176	2	31.7	0.08
48	3	–	–	–	0.169	1	48.0	0.06
64	3	–	–	–	0.171	0	64.0	0.05

C Reproducibility

Seed 20261020. Embedders are pinned by revision in `configs/pipeline.yaml`. Data lives under `data/external/` (git-ignored); aggregates in `results/wp03b_persian/`.

```
uv run hf auth login                                  # once, gated Khayyam
uv run python scripts/ext_download.py                 # 12 sources
uv run python scripts/s4_persian_cluster.py small # also: nano; nano --task text-matching
uv run python scripts/s4_cluster_plots.py             # curves, maps
uv run python scripts/s4_report_figs.py               # data overview, comparison, composition
```

```
uv run python scripts/s4_cluster_meaning.py       # label make-up per cluster
uv run python scripts/s4_chat_topics.py           # topic anchors
uv run python scripts/s4_chat_topics_fig.py
uv run python scripts/s4_latex_tables.py          # tables of this report
cd report/persian_clustering/latex && latexmk -pdf main.tex
```